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Data Science

Smart Student presence Monitoring System

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Project Guide
MS. Dipali ghat

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1. Introduction

- Problem Identified :
  - Manual attendance tracking and report generation is **time-consuming** and prone to errors.
  - Generating **Academics reports** and analyzing student progress manually is difficult.
- Solution Proposed :
  - Develop a **mobile-based system** to digitally record and monitor student attendance.
  - Provide a centralized platform for teachers to manage attendance and Academics for students to view records.

2. Objectives

1. To manage and store student registration details in a centralized system.
2. To monitor student attendance and send alerts when attendance falls below 75%.
3. To enable quick and convenient access to student records via mobile application
4. To generate essential report, such as attendance percentage , result summaries, and student lists, to support daily academic operation

3. Scope

1. Can be implemented in schools, colleges, and educational institutions.
2. Helps teachers manage attendance and academic records efficiently.
3. Allows students to view their attendance and performance reports.
4. Improves data analysis using graphical reports and dashboards.

4. Feature /Functionality

1. Feature 1: Attendance Management
   - Teachers can mark, edit, and update student attendance through the application.
2. Feature 2 :Marks Management
   - Teachers can record and update student marks for different subjects.
3. Feature 3 : Report Generation & Analytics
   - The system generates reports and displays charts or graphs to analyze attendance and academic performance.

5. Outcome of Project

1. Teachers can manage attendance digitally.
2. Student marks and records can be stored and updated easily.
3. Graphs and charts help in analyzing student performance trend.
4. Reduces manual work and improves academic monitoring.

6. Technology Stack

Mobile Application (Frontend)

- Flutter
- Single codebase for Android and iOS

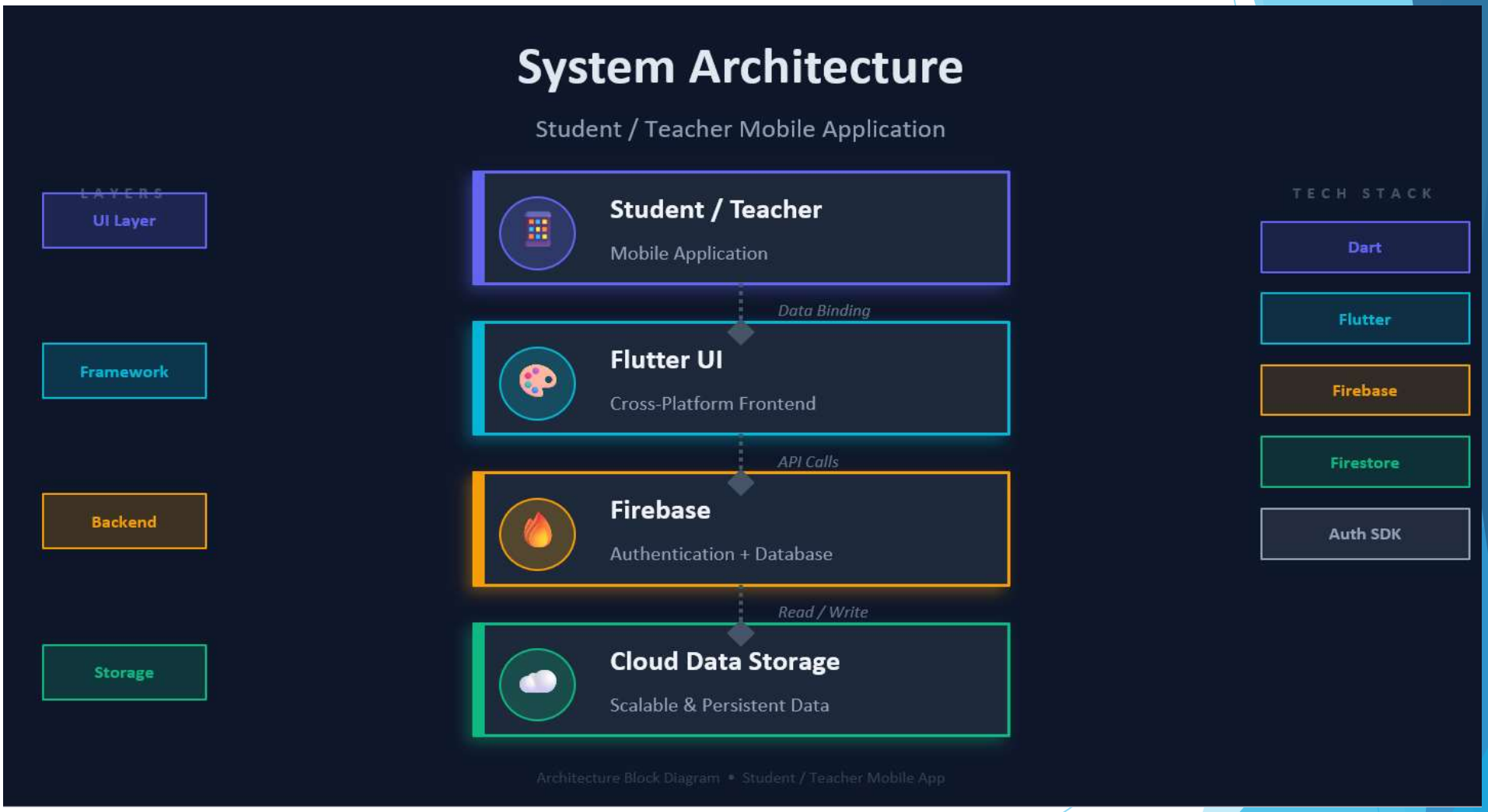
Backend

- Firebase
- Provides authentication and cloud database services
- Enables real-time data synchronization

Database

- Firebase
- Stores student details, attendance records, and marks securely

7. Block Diagram



Student / Teacher Mobile App

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Flutter UI

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Firebase

(Authentication + Database)

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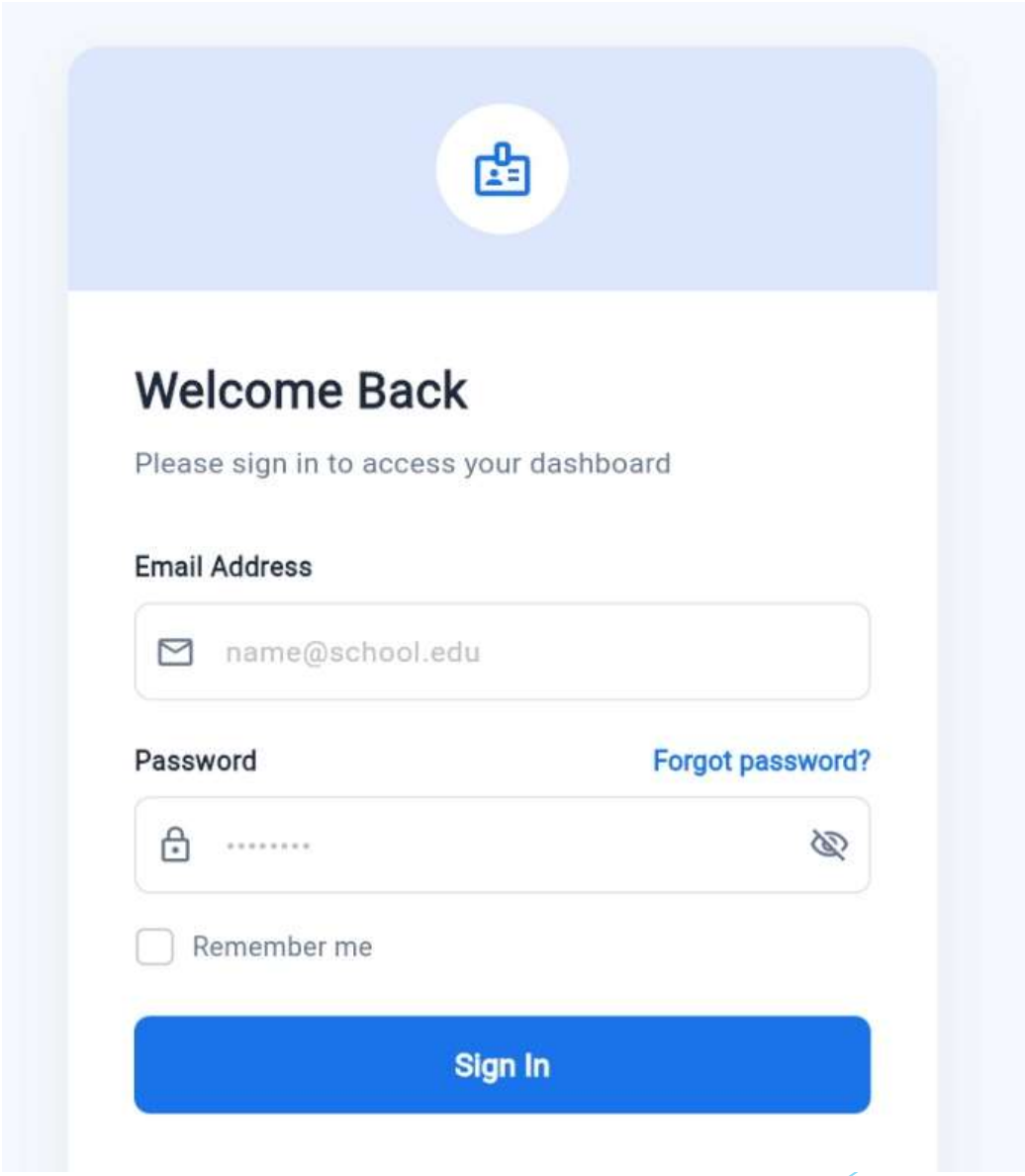
Cloud Data Storage

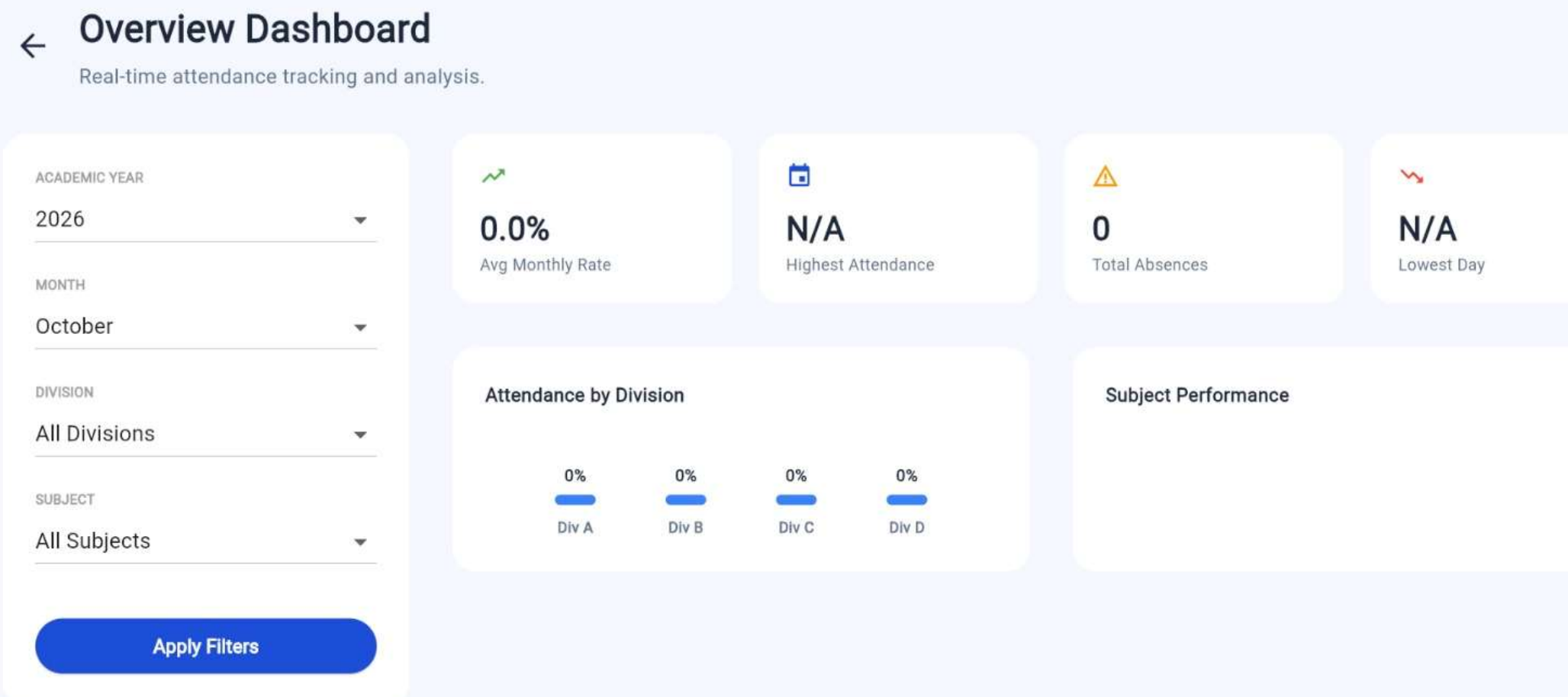
8. Sustainable Development Goal (SDG) Mapping

- **SDG 4 – Quality Education**
- This project supports quality education by providing a smart digital system for monitoring student attendance and academic performance. It helps teachers manage records efficiently and enables students to track their progress using reports and analytics. The system reduces manual work and improves academic monitoring.

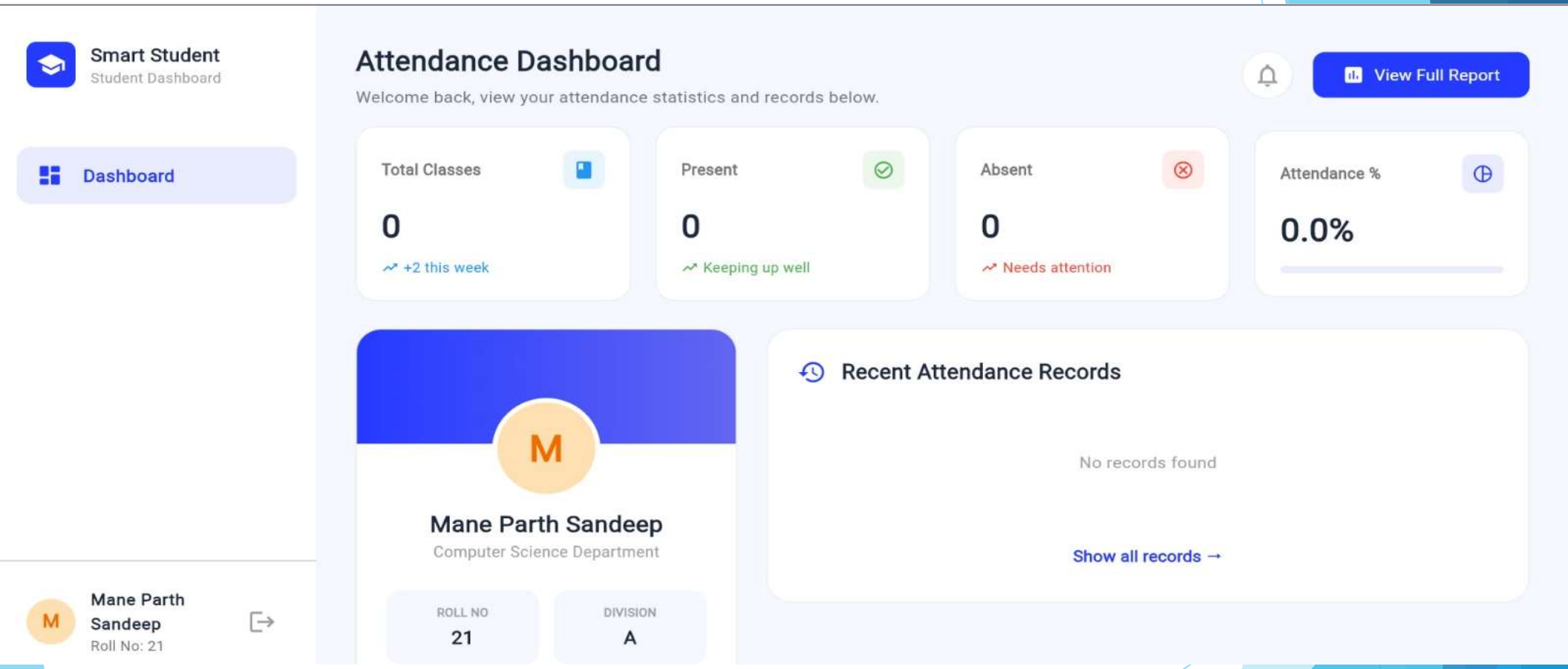
Frontend Implementation



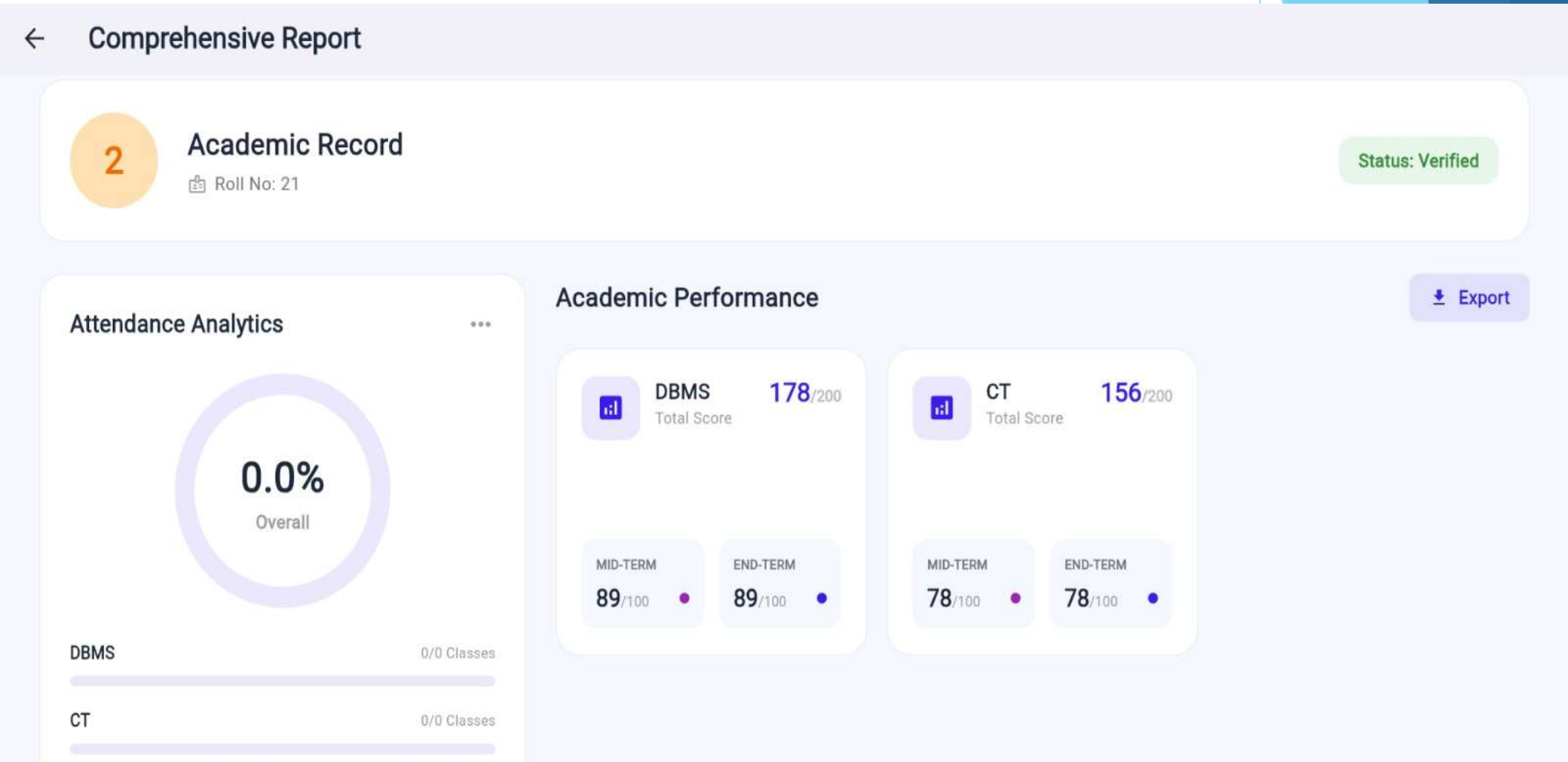
System Dashboard / Analytics



Student Attendance Output



Academic Report Output



Thank You...!!